

Note Selection:

Key Selection:

The key is the set of notes that can be randomly selected played by the plants. All of the plants will be selecting notes or a sequence of notes within this singular key determined by the time of the day and the weather.

The hour of the day determines what the key is {C, C#, D, Eb, E, F, F#, G, Ab, A, Bb, B}, weather it is raining or not determines whether that key is major or minor. The keys will be paired with hours of the day based on what feels right for the feeling of that time of day. Gmaj7 sounds really bright and a little sleepy so maybe that chord corresponds with 6:00pm, as the day's work is winding down and it's time to eat and relax. Or Dmaj7 which sounds a bit like it's leading into something else or anticipating something, maybe that could be paired with 10:00am as the day is getting busier and more hectic.

Additionally the CRTV's must only be on for part of the day. A reasonable time would be from 8am to 8pm as to maximize the amount of time people will be able to hear it playing music. This also works out well because there are 12 major and minor keys to choose from.

Octaves Selection:

Octaves are basically how high or low the notes are. The note letters themselves repeat 8 times, but the octave is which repeat of the notes you're using repeat[0-8]. The notes themselves on a typical piano go from A₀ to C₈. The subscript only increments up once it passes B into the next C. I.E. B₁ is followed by C₂. Not every octave is the best to listen to ambiently though, the very low octave and the very high octaves can be obnoxious. So it is best to stay relatively close to the center of the range. **It might be fun to add a way for the octave range to change, 1-3 and 3-5 would still be fine.** For now let's just say the note is fine to choose Octaves 2-4.

Example:

Time: 8:00am

Weather: Not Raining

Key: Cmaj7

Note Options: {C₂, E₂, G₂, B₂, C₃, E₃, G₃, B₃, C₄, E₄, G₄, B₅}

muted.io/piano-chords/

Note Playing:

Changes within the signals of the plants will determine when to play notes or possibly brief arpeggios. The signals from the plants can be either completely inactive or change too quickly to be coherent. So the average of the signals will be taken over a period of time to make the notes are played randomly but reasonably often.

We will be taking several measurements a second, the average will be found for each second. Creating one combined data point for each second. Then we will take the average of those second averages over some determined amount of time X. We will not be taking a running average we will be polling on what the average was over the last X seconds every X seconds. We will always keep track of 2 X second segment averages. The average of the last X seconds and the average of the previous X seconds before that.

We can do several things with these to determine weather and how to play our note/notes. We can take the difference between the two averages, depending on the magnitude of the difference we can decide weather to play a note or not.

Example:

Interval: 30 seconds

Previous average: 500

Current average: 5000

Difference Magnitude: 4500

Difference Direction: +

Note Play Qualifier: Difference $\geq$ 1000

We've been measuring for one minute, the average for the first 30 seconds was 500 and the average over the last 30 seconds was 5000, the difference is 4500. We decided to play a note if the difference was greater than or equal to 1000 so we will play a note.

We can add more modifiers than that if we want though. Maybe the Difference has to be 500 to play a note but if it's greater than 500 we can play an arpeggio or a chord to indicate the plant is even more active now. An arpeggio is a series of notes in rapid succession either ascending or descending, and a chord is just several notes played together. If we implemented this we would have to have something like a list of lists of our notes the arpeggios or chords would play. EX for Cmaj7 it would be $\{\{C_2, E_2, G_2, B_2\}, \{C_3, E_3, G_3, B_3\}, \{C_4, E_4, G_4, B_5\}\}$. Our chord would or arpeggio would just be a single one of these sets as having it be our whole octave range would likely be excessive. I think a fun idea would be use the direction of the difference to determine weather the arpeggio is ascending or descending. I.E. The average grew larger by 1000 so lets play it ascending, C_2, E_2, G_2, B_2 , or it shrank by 1000 so lets play it descending, B_2, G_2, E_2, C_2 . But if it just grew by like 100 let's just play a single note from our selection.

Differences in Plants:

Each plant will have varying levels of activity so it makes sense to change the average period over which they're being measured. This is not required but is likely a good idea. I would also be fun to have different plants be different instruments. For instruments it would be best to choose inoffensive instruments from high fidelity sound fonts. Like, piano, electric piano, rhodes piano, vibraphone, xylophone, bass guitar, bass synth. It's best to avoid brass or woodwinds as they can be shrill or generally annoying.

External Device Requirements:

We will have to be processing the signals of 6-8 plants concurrently. We will need to have some way to electronically trigger the automatic watering system. Likely just a bluetooth triggered powerstrip which turns off or on when given the signal. There would be a pre-selected setting on the watering system that would hopefully run when we give it power again. The pre-selected cycles are time based, like water for 10 minutes every 12 hours. If this is not possible with what we have this task will be given to the person who ends up watering after we're gone. Additionally we will need to turn off the CRTVs electronically as well. They cannot have their power cut as that would damage them so we must find a way to either physically press the button to turn them off or use a remote to send a signal to turn them off.

Good Soundfont choices:

The name in bold is the soundfont the name not in bold is the instrument, the #+ is the octaves that it sounds good in.

FluidR3 GM:

Yamaha Grand Piano 3+
Bright Yamaha Grand Piano 3+
Rhodes EP 2+
Vibraphone 2+
Marimba 2+
Jazz Guitar 1-4
Synth Strings 1 2-4

MuseScore_General:

Grand Piano 2+
Bright Grand Piano 1+
Vibraphone 2+
Marimba 3+
Nylon String Guitar 2+

Jazz Guitar +1